

Project Decisions & Architecture Report

Project: Stock Market Trend Analysis — next-day log-return forecasting **Tickers:** `^GSPC`, `AAPL`, `AMZN`, `NVDA` · **Window:** 2016-05-23 → 2026-05-20 (pinned, 2,513 rows/ticker) **Owner:** Ali Agela · **Seed:** 42 · **Status:** M1 + M2 + M3 + M3.5 + M4 complete, all multi-agent-audited. The attention LSTM was trained on a Colab GPU (2026-07-06); its predictions are in `holdout_predictions.parquet`. The exec-summary table below and the whole project are current as of the M4 evaluation. **Sources:** `Plans/progress_checklist.md` (Decisions Log), `reports/milestones/M1-M3.md`, `Plans/milestone1-4_*.md`, the three notebooks, `data/processed/feature_roles.json`, `feature_dictionary.md`, `data/raw/snapshot_hashes.json`

1. Executive summary

This capstone asks a deliberately falsifiable question: *given 10 years of daily price data for the S&P 500 index and three large-cap tech stocks, can we forecast the next-day log return better than a naive baseline?* The pipeline spans four milestone specs — data collection and preprocessing (M1), exploratory data analysis (M2), model building (M3), and evaluation/presentation (M4) — executed as Jupyter `.py` notebooks synced to `.ipynb` via `scripts/py_to_nb.py`, snapshotted to committed CSVs for reproducibility, and reviewed at every milestone by independent expert agents.

The honest headline result is that **no model demonstrates economically usable out-of-sample skill**. On the sealed holdout (2025-01-02 → 2026-05-20, opened exactly once), pooled across tickers:

Model	Holdout RMSE	vs naive (Diebold-Mariano p)	DirAcc	DirAcc p
<code>naive_zero</code>	0.02109	n/a	n/a	n/a
ARIMA	0.02107	0.783 (tie)	0.514	0.294 (coin flip)
GJR-GARCH-t (mean)	0.02106	0.589 (tie)	0.543	0.002 (significant)
LightGBM (baseline, un-tuned)	0.02168	0.003 (WORSE)	0.483	0.226 (coin flip)
LightGBM (tuned, M3.5)	0.02107	0.533 (tie)	0.542	0.002 (significant)
LSTM+Attention (real GPU)	0.02031	0.004 (WORSE)	0.491	0.574 (coin flip)

Updated after the M3.5 hardening pass and the real Colab GPU run (2026-07-06). ARIMA ties naive on RMSE and is a directional coin flip. The un-tuned LightGBM overfits (holdout worse than naive, DM p = 0.003). The M3.5 Optuna tuning on a purged walk-forward CV fixed that overfit: the tuned LightGBM was promoted because it went from worse-than-naive to naive-tying (DM vs baseline p = 0.0025). Two models keep a statistically significant directional edge, GJR-GARCH at 0.543 and the tuned LightGBM at 0.542 (both p = 0.002), but both predict up on 99 to 100 percent of days, so that accuracy is the holdout up-day rate (0.542), the market's drift in a calm period rather than timing. The attention LSTM edge did not reproduce on the real GPU run (0.542 on an earlier forced-CPU run, 0.491 on the GPU run, a coin flip). No model beats naive on

RMSE, and the one edge that survives is drift that transaction costs erase (an illustrative $\wedge$ GSPC timer switches 174 times and collapses to Sharpe 0.33 at 10 bps, below buy-and-hold's 0.99, daily returns not significant at $t = 1.38$, $p = 0.168$). The efficient-market ceiling holds, and this negative result is the deliverable. See [reports/milestones/M4.md](#) for the full evaluation.

2. Milestone-by-milestone decisions

M1 — Data collection & preprocessing (*complete; self-audit 19/19*)

What was decided. Forecast **log return**, not **Close**; source from **Yahoo Finance** via **yfinance** with a **direct chart-API fallback**, snapshotted to committed CSVs; **retain and flag** outliers rather than cap them; split **chronologically** (train $\leq$ 2023-12-29, val = 2024, holdout = 2025+); build a **next-day target** and an explicit 30→39-column **MODEL_FEATURES** set ([feature_roles.json](#)).

Why (from the Decisions Log). ADF confirms **Close** is non-stationary ($p \geq 0.86$) while **log_return** is stationary ($p = 0.0000$), so ARIMA can fit returns with **d = 0** — "standard finance practice." A random split would leak the future; a chronological split with holdout = 2025+ reserves the AI cycle for an unbiased final test. **yfinance** + a committed CSV snapshot is the "reproducibility anchor: re-runs read from CSV not the live API." Outliers encode "real regime events — capping would destroy the signal we need to learn." The direct [query1.finance.yahoo.com](#) chart API is a fallback because the consent endpoint [fc.yahoo.com](#) was blocked from the local sandbox.

What was produced. 2,513 rows/ticker over 2016-05-23 → 2026-05-20 (SHA256-recorded in [snapshot_hashes.json](#)); a 55-column leakage-safe frame written to [train_fe.parquet](#) (7,660 rows), [val_fe.parquet](#) (1,008 rows), [holdout_fe.parquet](#) (1,384 rows); plus [feature_dictionary.md](#), [feature_roles.json](#), and [feature_nan_report.md](#).

Key metrics. ADF **log_return** $p = 0.0000$ all tickers; **Adj Close** $p = 0.86\text{--}0.998$. Ljung-Box(r^2) lag-1 $p \approx 0$ all with ACF(r^2) up to 0.453 ($\wedge$ GSPC) — i.e. **stationary in mean but conditionally heteroskedastic**. Log-return kurtosis 5.1–16.8 ($\wedge$ GSPC fattest, COVID-driven; ex-COVID 6.46). Rolling-MAD tail-event flag rate 5.85%–7.16%.

M2 — Exploratory data analysis (*complete; self-audit 13/13, 18 figures, 5 hypotheses*)

What was decided. EDA reads [train_fe.parquet](#) **only** (val/holdout sealed); rank features by **mutual information**, not just Pearson; test the **January effect per ticker**, not pooled; add a **leverage-effect test**; recommend **asymmetric GARCH (EGARCH/GJR)** for the M3 variance model.

Why (from the Decisions Log). Pearson is ~ 0 for every feature and "hid the signal"; MI shows the volatility features carry $\sim 67\times$ the noise floor. Pooling four tickers correlated at ~ 0.65 "inflated effective-n $\sim 3\times$," masking AMZN's January significance. Down days raise next-day volatility more, so "symmetric GARCH would miss the asymmetry."

What was produced. 18 figures ([M2_fig01..18](#)), including the MI ranking (Fig 17) and leverage test (Fig 18); five hypotheses with verdicts; a modelling-implications hand-off.

Key metrics. Annualised vol $\wedge$ GSPC 18.8% → NVDA 49.6% (max drawdown $-33.9\% \rightarrow -66.3\%$). Cross-ticker correlation $\wedge$ GSPC–AAPL 0.78 (range 0.58–0.78). ACF(return) lag-1 = -0.162 ; ACF(return²) lag-1 = $+0.487$. MI: [parkinson_vol_21](#) 0.160 vs shuffle floor 0.0024 ($\sim 67\times$) vs its own Pearson ~ 0.01 . Hypotheses: H1 normality

REJECTED (JB 2,129–21,958, $p \approx 0$); H2 ARCH **CONFIRMED** (LM 65–757, $p \approx 0$); H3 tech corr > 0.6 **YES**; H4 January significant **AMZN only** ($t = 2.24$, $p = 0.027$; the invalid pooled test gave $p = 0.103$); H5 leverage **ASYMMETRIC all** ($p = 7e-6 \dots 0.035$).

M3 — Model building (*core complete; self-audit 19/19 post-enhancement; LSTM full-train pending Colab*)

What was decided. Score **baselines first**, then per-ticker **ARIMA with order selected by BIC**, per-ticker **GJR-GARCH(1,1) Student-t** (evaluated as a variance model), a **global attention-LSTM** (Colab), and — in the enhancement pass — **LightGBM on 39 features + 9 exogenous macro series** with a z-scored directional ensemble; the **holdout is opened exactly once**.

Why (from the Decisions Log). BIC over AIC because "AIC over-selected (2,0,2); BIC gives parsimonious (1,0,0)/(0,0,1)." GJR-GARCH because M2 proved the leverage effect. Significance tests (binomial + Diebold-Mariano) because "a 55% DirAcc read as skill without a test." Exogenous data is "the only break from price-history-only," fetched leakage-safe (market-priced daily, known at close t , no CPI-style release lag). LightGBM was left "deliberately un-tuned ... so the overfit is the lesson."

What was produced. `models/{arima,garch}*.pkl`, `lgbm_global{,_final}.txt`, `lstm_attention_smoke.keras`, `val_scores.csv`, and `holdout_predictions.parquet` (with `y_pred_arima`, `y_pred_garch`, `y_pred_lgb`, `y_pred_lstm` placeholder).

Key metrics. Baselines (val): `naive_zero` RMSE 0.02041; `persistence` DirAcc 0.512; `moving_avg_20` 0.520. ARIMA orders: ^GSPC/AMZN/NVDA (1,0,0), AAPL (0,0,1). GJR-GARCH $\gamma > 0$ all (^GSPC 0.233), $\nu \approx 5$. Val leader was GJR-GARCH-mean (DirAcc 0.555, $p = 0.0006$) — but selection-biased, so only the holdout (Section 1) counts.

Enhancement pass (2026-06-17)

Added the nine exogenous features (`vix_level`, `vix_log_change`, `vix_z_60`, `is_high_vol_regime`, `tnx_level`, `tnx_change`, `term_spread`, `term_spread_change`, `dxy_log_return`), LightGBM, a Bahdanau-style attention layer on the LSTM, and a z-scored ensemble. **The instructive finding is the overfit:** exogenous data is genuinely informative in-sample (8 of LightGBM's top-15 importances are exogenous) and lifts val DirAcc to 0.539 ($p = 0.015$), but does **not** generalize — holdout RMSE 0.02165 > naive, DM $p = 0.003$, DirAcc 0.499. More data and more model capacity bought no out-of-sample lift, sharpening the EMH conclusion.

3. The multi-agent audit process

Every milestone was reviewed by **independent expert agents** — an **Investment Researcher** and a **Model QA Specialist** on all three, plus a **Data Engineer** on M1 — who reproduced each headline number before flagging issues by severity. This adversarial discipline is a core part of the story: several audit findings were BLOCKERS that invalidated the stated task until fixed.

M1 (three agents). Material findings and fixes:

- **BLOCKER — no next-day target / off-by-one leak (Model QA):** the target was the same-row `log_return`, making the stated "next-day forecast" unbuildable and same-day columns leaky. Fix:

`target_log_return = groupby(ticker).log_return.shift(-1)`, built per split, verified
`target[t] == log_return[t+1]`.

- **BLOCKER — non-deterministic date window (Data Eng + Model QA):** a `today`-anchored window made a fresh clone/Colab non-reproducible. Fix: pinned `START=2016-05-23`, `END=2026-05-21` (exclusive), recorded in `snapshot_hashes.json`.
- **BLOCKER — Volume dtype drift (Data Eng + Model QA):** `yfinance` returned `int64`, the fallback `float64`, the dictionary claimed `float64`. Fix: a single `enforce_schema()` on all fetch paths + post-concat assertion.
- **HIGH — global-mean target-encoding leakage (M1 v2 audit):** `ticker_mean_log_return` was a global `groupby(ticker).mean()`, so every early-train row saw the whole-train mean. Fix: `expanding(min_periods=20).mean().shift(1)` (~1,894 unique values/ticker, was 1).
- **HIGH — IQR outliers over-flagged NVDA (M1 v2 audit):** fixed-quantile IQR flagged 14% of NVDA. Fix: rolling 60-day MAD, threshold 3.5 (flag rates 5.9%–7.2%).
- **HIGH — "stationary" overclaim (Investment Researcher):** returns are conditionally heteroskedastic. Fix: added Ljung-Box(r^2) + ACF(r^2), reworded to "stationary in mean," flagged ARIMA+GARCH for M3.
- Plus: `MODEL_FEATURES/feature_roles.json` role separation, `is_warmup` flag + NaN report, tail-event renaming with NaN warm-up, Colab bootstrap, and partial-ticker-failure handling.

M2 (two agents). `pooled January invalid` → `per-ticker` (unmasked AMZN $p = 0.027$); `Pearson hides the signal` → `MI ranking` added (Fig 17); `vacuous sealing check` → `real read_parquet-path tracking`; leverage test added (H5/Fig 18) driving the EGARCH/GJR recommendation; ACF lag-1 reframed as negative microstructure; returns-based decomposition cross-check.

M3 (two agents). `off-by-one in the ARIMA one-step (BLOCKER)` — the forecast was taken before appending today's return, predicting the same day but scored against the next (val ARIMA DirAcc 0.536 → 0.521, the old edge was the artifact; holdout RMSE 0.02129 → 0.02107); `BIC not AIC`; binomial + Diebold-Mariano significance tests added; GARCH evaluated by QLIKE + Mincer-Zarnowitz on variance (not point RMSE) and persisted via `pickle`; costed backtest with Sharpe/maxDD/t-stat. **Enhancement re-audit:** confirmed exogenous leakage-clean (VIX same-day corr -0.70 vs next-day $+0.05$ = no label contamination), **re-tested GARCH out-of-sample** (yielding the 54.3% $p = 0.002$ survivor), and corrected the **"EGARCH" mislabel to GJR-GARCH ($\alpha=1$)** everywhere, softened the VIX release-timing wording, and documented LightGBM as deliberately un-tuned.

4. Leakage-prevention invariants

The pipeline enforces five guards, each agent-verified:

1. **Expanding-mean target encoding.** `ticker_expanding_mean/std = expanding(min_periods=20).{mean,std}().shift(1)` — each row sees only its strict past (`np.allclose` reproduced independently by the agents).
2. **.shift(1).rolling(...)** on every rolling/lag feature. SMAs, realized vols, `rolling_std_20`, Parkinson vol, RSI, `volume_z_20`, and all lags shift *before* rolling — no same-day value enters its own window.
3. **Next-day target built per split.** `target_log_return = groupby(ticker).log_return.shift(-1)` is computed *after* splitting, so the last row of each split per ticker is NaN (4 NaN targets/split = 1/ticker) — no label leaks across a boundary.

4. **Fit on train only.** The LSTM `StandardScaler` is fit on train exclusively; GARCH uses `last_obs = first_val` so parameters come only from train; ARIMA one-step is causal (append-then-forecast); the target is never in the feature matrix.
5. **Holdout opened exactly once**, in the final M3 step; a logged `read_parquet` wrapper + self-audit check *prove* M2 touched only `train_fe.parquet`.

Exogenous checks (enhancement pass). VIX matches the raw snapshot exactly; same-day VIX correlation -0.70 vs next-day $+0.05$ proves no label contamination; forward-fill is backward-safe and ticker-scoped; `term_spread` (10y – 3m) is economically correct, not a unit artifact; the attention layer pools over the time axis correctly; all features are market-priced daily and known at close t (no release-lag lookahead).

5. Key quantitative findings

Significance-tested val→holdout skill decay is the central quantitative story: the only significant val edge (GJR-GARCH-mean DirAcc 0.555, $p = 0.0006$) was selected on val (selection bias), and the sealed holdout shrinks it to a marginal 0.543 ($p = 0.002$) while ARIMA (0.521 → 0.514) and LightGBM (0.539 → 0.499) decay to coin flips. Every model ties naive on RMSE (DM $p \geq 0.07$ on val; ≥ 0.59 on holdout except LightGBM, which is *worse*).

Volatility clustering is confirmed everywhere: ARCH-LM $p \approx 0$ for all four tickers, Ljung-Box(r^2) $p \approx 0$, and ACF(r^2) lag-1 = 0.487 ($\wedge$ GSPC) — the single most important result for model selection. **The leverage effect** is significant for all four (down-day next-day vol > up-day, p 7e-6 ... 0.035), and the fitted GJR-GARCH recovers it internally ($\gamma > 0$ all; $\wedge$ GSPC 0.233).

MI vs Pearson quantifies why a nonlinear model was warranted at all: `parkinson_vol_21` carries MI 0.160 ($\sim 67\times$ the 0.0024 noise floor, $\sim 3\times$ the best linear feature) while its Pearson is ~ 0.01 — real, mostly-nonlinear, volatility-based structure invisible to a linear screen. **Fat tails** are pervasive (log-return kurtosis 5.1–16.8; GARCH $v \approx 5$), justifying Student-t errors over Gaussian.

Taken together, these are consistent, textbook **EMH evidence**: next-day *mean* is near-unpredictable while next-day *variance* is strongly predictable — and even the predictable variance, converted to a directional signal, is uneconomic after costs.

6. Limitations & outstanding work

- **LSTM trained on Colab GPU (2026-07-06).** The full 2-layer/64-unit/20-epoch attention-LSTM ran; its holdout directional accuracy is 0.491 ($p = 0.574$, a coin flip), and the earlier forced-CPU 0.542 did not reproduce. So the deep arm exists and its honest verdict is in.
- **M4 complete** (11/11): final performance table vs a perfect-foresight bound, top-10-worst error narratives, confusion matrix, ethics/misuse audit with the explicit "not investment advice" disclaimer, and a 12-slide HTML deck are all delivered. See `reports/milestones/M4.md` and `reports/M4_presentation.html`.
- **Finance-specific limits.** The EMH bounds predictability from prices alone; the holdout is a **single calm regime** (2025–26, lower vol than train), so no claim generalizes across bull/bear; transaction costs erase the marginal GARCH edge; the four tickers are **survivors** (survivorship bias) and $\wedge$ GSPC is not directly tradeable; daily granularity hides intraday signal; multiple comparisons (4 tickers $\times$ orders $\times$ models) mean any single "win" should be read skeptically. The GARCH variance was scored against a noisy

squared-return proxy; M4 could add a range-based realized variance and a VaR/Kupiec exceedance backtest.